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FACULTY OF INFORMATION TECHNOLOGY

Student Management System

Project Information

Project Name	Students Management System
Course Title	Introduction To Computing – LAB (1 ST Semester)
Submission Date	17-June-2025
Faculty	Faculty Of Information Technology
Submitted By	Sultan Shahzad Awan



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Program Description

The Student Management System is a comprehensive C++ application designed to manage student records for an educational institution. The system allows for:

- 1) Student registration with detailed personal and academic information
- 2) Administrative functions to view, search, update, and remove student records
- 3) Support for different academic levels (Undergraduate, Graduate, Postgraduate)
- 4) Multiple program options (AI, CS, SE, DS)
- 5) Automated scholarship calculations based on academic performance
- 6) Hostel and transport facility management

Key Features

1. Multi-Level Student Management:

- ✚ Undergraduate students (with intermediate marks tracking)
- ✚ Graduate students (with GPA tracking)
- ✚ Postgraduate students (with GPA tracking)

2. Comprehensive Student Information:

- ✚ Personal details (name, father's name, CNIC, gender, age, address)
- ✚ Contact information
- ✚ Academic details (program, level, department, semester)
- ✚ Performance metrics (marks or GPA)
- ✚ Facility requirements (hostel, transport)

3. Administrative Functions:

- ✚ View all student records in tabular format
- ✚ Search students by name
- ✚ Update student information



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4. Input Validation:

- ✚ Comprehensive validation for **all input fields**
- ✚ Case-insensitive input handling
- ✚ Proper data type verification

5. User-Friendly Interface:

- ✚ Menu-driven navigation
- ✚ Clear prompts and feedback
- ✚ `System("cls")` for clean screen transitions

Features Used

1. Core C++ Features:

- ✚ Arrays for data storage
- ✚ Strings for text handling
- ✚ Control structures (**loops**, **conditionals**)
- ✚ Function (**Setw**)

2. Standard Libraries:

- ✚ `<iostream>` for **input/output operations**
- ✚ `<string>` for **string manipulation**
- ✚ `<iomanip>` for **formatted output**

3. System Integration:

- ✚ `system("cls")` for screen clearing



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Pseudo Code

BEGIN PROGRAM

DECLARE constant MAX = 100

DECLARE arrays for student attributes (name, fatherName, cnic, etc.)

DECLARE totalStudents = 0, idCounter = 1

WHILE true DO

DISPLAY main menu

GET user choice

IF choice is "Student" THEN

DISPLAY level selection menu

GET level choice

IF level is "Undergraduate" THEN

DISPLAY program selection

GET program choice

COLLECT student information with validation:

- Name (alphabets only)
- Father's name (alphabets only)
- CNIC (13 digits)
- Gender (M/F)
- Address



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- Age (**positive number**)
- Contact (**11 digits**)
- Intermediate marks (**0-1200**)
- Hostel requirement (**Y/N**)
- Transport requirement (**Y/N**)

CALCULATE scholarship based on marks

INCREMENT totalStudents

ELSE IF level is "Graduate" or "Postgraduate" **THEN**

DISPLAY program selection

GET program choice

COLLECT student information with validation:

- Similar to undergraduate but with CGPA (0.0-4.0) instead of marks

CALCULATE scholarship based on CGPA

INCREMENT totalStudents

ELSE IF choice is "Back" **THEN**

RETURN to main menu

END IF

ELSE IF choice is "Admin" **THEN**

PROMPT for admin credentials



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IF credentials valid THEN

WHILE true DO

DISPLAY admin menu

GET admin choice

IF choice is "Display All Students" THEN

PRINT all students in formatted table

SHOW scholarship status

ELSE IF choice is "Search by Name" THEN

GET search name

FIND and DISPLAY matching student

ELSE IF choice is "Update Student" THEN

GET student ID

DISPLAY current information

GET field to update

VALIDATE and APPLY update

ELSE IF choice is "Remove Student" THEN

GET student ID

CONFIRM removal



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REMOVE student if confirmed

ELSE IF choice is "Back" **THEN**

RETURN to main menu

END IF

END WHILE

ELSE

DISPLAY invalid credentials message

END IF

ELSE IF choice is "Exit" **THEN**

EXIT program

END IF

END WHILE

END PROGRAM

Code On Visual Studio :

```
#include <iostream>
#include <string>
#include <iomanip> // For setw
using namespace std;
```




```
int main() {
const int MAX = 100;
string name[MAX], fatherName[MAX], cnic[MAX], gender[MAX];
string address[MAX], contact[MAX], field[MAX], level[MAX], program[MAX];
int age[MAX], interMarks[MAX];
string semester[MAX], department[MAX];
float gpa[MAX];
int studentID[MAX];
string hostel[MAX], transport[MAX];

int totalStudents = 0;
int idCounter = 1;

while (true) {
system("cls");
cout << "\n\t\t\t\t\t\t\t\t\t\t===== STUDENT MANAGEMENT SYSTEM =====>> endl;
cout << "\n\n\n-----MAIN MENU -----&#x2D;\n";
cout << "1. Student\n\n2. Admin\n\n3. Exit\n\n";
cout << "Enter your choice: ";
string input;
getline(cin, input);

// Convert input to lowercase for case-insensitive comparison
for (int i = 0; input[i]; i++) {
if (input[i] >= 'A' && input[i] <= 'Z') {
input[i] = input[i] + 32;
}
}

if (input == "1" || input == "s" || input == "student") {
while (true) {
system("cls");
cout << "\n\n\n-- Student Level Selection --\n\n";
cout << "1. Undergraduate\n\n2. Graduate\n\n3. Postgraduate\n\n4. Back\n\n";
cout << "Enter choice: ";
string levelChoice;
getline(cin, levelChoice);

// Convert to lowercase
for (int i = 0; levelChoice[i]; i++) {
if (levelChoice[i] >= 'A' && levelChoice[i] <= 'Z') {
levelChoice[i] += 32;
}
}

if (levelChoice == "1" || levelChoice == "u" || levelChoice == "undergraduate") {
string selectedLevel = "Undergraduate";
while (true) {
system("cls");
cout << "\n\n\n-- Program Selection --\n\n";
cout << "1. AI\n\n2. CS\n\n3. SE\n\n4. DS\n\n5. Back\n\n";
cout << "Enter choice: ";
string programChoice;
```



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```
getline(cin, programChoice);
```

```
// Convert to lowercase
for (int i = 0; programChoice[i]; i++) {
    if (programChoice[i] >= 'A' && programChoice[i] <= 'Z') {
        programChoice[i] += 32;
    }
}

string selectedProgram;
if (programChoice == "1" || programChoice == "ai") {
    selectedProgram = "AI";
}
else if (programChoice == "2" || programChoice == "cs") {
    selectedProgram = "CS";
}
else if (programChoice == "3" || programChoice == "se") {
    selectedProgram = "SE";
}
else if (programChoice == "4" || programChoice == "ds") {
    selectedProgram = "DS";
}
else if (programChoice == "5" || programChoice == "b" || programChoice == "back") {
    break;
}
else {
    cout << "\nInvalid choice. Press Enter to try again...";
    cin.ignore();
    continue;
}

system("cls");
cout << "\n\n--- Enter Student Information ---\n";

// Name (with validation)
while (true) {
    cout << "\nName:";
    getline(cin, name[totalStudents]);
    bool valid = true;
    for (int i = 0; name[totalStudents][i]; i++) {
        char c = name[totalStudents][i];
        if (!(c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z') || c == ' ')) {
            valid = false;
            break;
        }
    }
    if (valid && name[totalStudents][0] != '\0') {
        break;
    }
    cout << "\nInvalid name. Only alphabets and spaces allowed. Press Enter to try again...";
    cin.ignore();
    system("cls");
    cout << "\n\n--- Enter Student Information ---\n\n";
}
```



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```
// Father's Name (with validation)
while (true) {
    cout << "\nFather's Name: ";
    getline(cin, fatherName[totalStudents]);
    bool valid = true;
    for (int i = 0; fatherName[totalStudents][i]; i++) {
        char c = fatherName[totalStudents][i];
        if (!(c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z') || c == ' ')) {
            valid = false;
            break;
        }
    }
    if (valid && fatherName[totalStudents][0] != '\0') {
        break;
    }
    cout << "\nInvalid father's name. Only alphabets and spaces allowed. Press Enter to try again...";
    cin.ignore();
    system("cls");
    cout << "\n\n--- Enter Student Information ---\n";
    cout << "\nName: " << name[totalStudents] << endl;
}

// CNIC (with validation)
while (true) {
    cout << "\nCNIC (13 digits without dashes): ";
    getline(cin, cnic[totalStudents]);
    bool valid = true;
    int digits = 0;
    for (int i = 0; cnic[totalStudents][i]; i++) {
        if (cnic[totalStudents][i] < '0' || cnic[totalStudents][i] > '9') {
            valid = false;
            break;
        }
    }
    digits++;
    if (valid && digits == 13) {
        break;
    }
    cout << "\nInvalid CNIC. Must be exactly 13 digits. Press Enter to try again...";
    cin.ignore();
    system("cls");
    cout << "\n\n--- Enter Student Information ---\n";
    cout << "\nName: " << name[totalStudents] << endl;
    cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
}

// Gender (with validation)
while (true) {
    cout << "\nGender (M/F): ";
    getline(cin, gender[totalStudents]);
    if (gender[totalStudents][0] && !gender[totalStudents][1]) {
        char g = gender[totalStudents][0];
        if (g >= 'A' && g <= 'Z') {
```



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```
g += 32;
}
if (g == 'm') {
gender[totalStudents] = "Male";
break;
}
else if (g == 'f') {
gender[totalStudents] = "Female";
break;
}
}
cout << "\nInvalid gender input. Press Enter to try again...";
cin.ignore();
system("cls");
cout << "\n--- Enter Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
cout << "\nCNIC: " << cnic[totalStudents] << endl;
}

// Address
cout << "\nAddress: ";
getline(cin, address[totalStudents]);

// Age (with validation)
while (true) {
cout << "\nAge: ";
string ageInput;
getline(cin, ageInput);
bool valid = true;
for (int i = 0; ageInput[i]; i++) {
if (ageInput[i] < '0' || ageInput[i] > '9') {
valid = false;
break;
}
}
if (valid && ageInput[0] != '\0') {
age[totalStudents] = 0;
for (int i = 0; ageInput[i]; i++) {
age[totalStudents] = age[totalStudents] * 10 + (ageInput[i] - '0');
}
if (age[totalStudents] > 0) break;
}
cout << "\nInvalid age. Please enter a positive number. Press Enter to continue...";
cin.ignore();
system("cls");
cout << "\n--- Enter Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
cout << "\nCNIC: " << cnic[totalStudents] << endl;
cout << "\nGender: " << gender[totalStudents] << endl;
cout << "\nAddress: " << address[totalStudents] << endl;
}
```



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```
// Contact (with validation)
while (true) {
    cout << "\nContact Number (11 digits): ";
    getline(cin, contact[totalStudents]);
    bool valid = true;
    int digits = 0;
    for (int i = 0; contact[totalStudents][i]; i++) {
        if (contact[totalStudents][i] < '0' || contact[totalStudents][i] > '9') {
            valid = false;
            break;
        }
        digits++;
    }
    if (valid && digits == 11) {
        break;
    }
    cout << "\nInvalid contact number. Must be exactly 11 digits. Press Enter to try again...";
    cin.ignore();
    system("cls");
    cout << "\n--- Enter Student Information ---\n";
    cout << "\nName: " << name[totalStudents] << endl;
    cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
    cout << "\nCNIC: " << cnic[totalStudents] << endl;
    cout << "\nGender: " << gender[totalStudents] << endl;
    cout << "\nAddress: " << address[totalStudents] << endl;
    cout << "\nAge: " << age[totalStudents] << endl;
}

// Intermediate Marks (with validation)
while (true) {
    cout << "\nIntermediate Marks (out of 1200): ";
    string marksInput;
    getline(cin, marksInput);
    bool valid = true;
    for (int i = 0; marksInput[i]; i++) {
        if (marksInput[i] < '0' || marksInput[i] > '9') {
            valid = false;
            break;
        }
    }
    if (valid && marksInput[0] != '\0') {
        interMarks[totalStudents] = 0;
        for (int i = 0; marksInput[i]; i++) {
            interMarks[totalStudents] = interMarks[totalStudents] * 10 + (marksInput[i] - '0');
        }
        if (interMarks[totalStudents] >= 0 && interMarks[totalStudents] <= 1200) {
            gpa[totalStudents] = -1;
            break;
        }
        else if (interMarks[totalStudents] > 1200) {
            cout << "\nObtained marks cannot be greater than total marks (1200). ";
        }
    }
    cout << "\nInvalid marks. Press Enter to try again...";
}
```



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```
cin.ignore();
system("cls");
cout << "\n--- Enter Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
cout << "\nCNIC: " << cnic[totalStudents] << endl;
cout << "\nGender: " << gender[totalStudents] << endl;
cout << "\nAddress: " << address[totalStudents] << endl;
cout << "\nAge: " << age[totalStudents] << endl;
cout << "\nContact: " << contact[totalStudents] << endl;
}

// Hostel (with validation)
while (true) {
cout << "\nRequire Hostel (Y/N/1/0): ";
getline(cin, hostel[totalStudents]);
if (hostel[totalStudents][0] && !hostel[totalStudents][1]) {
char h = hostel[totalStudents][0];
if (h >= 'A' && h <= 'Z') {
h += 32;
}
if (h == 'y' || h == '1' || h == 'yes') {
hostel[totalStudents] = "Yes";
break;
}
else if (h == 'n' || h == '0' || h == 'no') {
hostel[totalStudents] = "No";
break;
}
}
cout << "Invalid input. Please enter Y/N or 1/0. Press Enter to try again...";
cin.ignore();
system("cls");
cout << "\n--- Enter Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
cout << "\nCNIC: " << cnic[totalStudents] << endl;
cout << "\nGender: " << gender[totalStudents] << endl;
cout << "\nAddress: " << address[totalStudents] << endl;
cout << "\nAge: " << age[totalStudents] << endl;
cout << "\nContact: " << contact[totalStudents] << endl;
cout << "\nIntermediate Marks: " << interMarks[totalStudents] << "/1200" << endl;
}

// Transport (with validation)
while (true) {
cout << "\nRequire Transport (Y/N/1/0): ";
getline(cin, transport[totalStudents]);
if (transport[totalStudents][0] && !transport[totalStudents][1]) {
char t = transport[totalStudents][0];
if (t >= 'A' && t <= 'Z') {
t += 32;
}
if (t == 'y' || t == '1') {
```



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```
transport[totalStudents] = "Yes";
break;
}
else if (t == 'n' || t == '0') {
transport[totalStudents] = "No";
break;
}
}
cout << "\nInvalid input. Please enter Y/N or 1/0. Press Enter to try again...";
cin.ignore();
system("cls");
cout << "\n--- Enter Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
cout << "\nCNIC: " << cnic[totalStudents] << endl;
cout << "\nGender: " << gender[totalStudents] << endl;
cout << "\nAddress: " << address[totalStudents] << endl;
cout << "\nAge: " << age[totalStudents] << endl;
cout << "\nContact: " << contact[totalStudents] << endl;
cout << "\nIntermediate Marks: " << interMarks[totalStudents] << "/1200" << endl;
cout << "\nHostel: " << hostel[totalStudents] << endl;
}

// Assign remaining fields
studentID[totalStudents] = idCounter++;
program[totalStudents] = selectedProgram;
level[totalStudents] = selectedLevel;
semester[totalStudents] = "1st";
department[totalStudents] = "FOIT";
field[totalStudents] = selectedProgram;

totalStudents++;

cout << "\nStudent added successfully!\n";
cout << "\nPress Enter to return to Main Menu...";
cin.ignore();
break;
}
break;
}
else if (levelChoice == "2" || levelChoice == "g" || levelChoice == "graduate") {
string selectedLevel = "Graduate";
while (true) {
system("cls");
cout << "\n\n-- Graduate Program Selection --\n\n";
cout << "1. AI\n\n2. CS\n\n3. SE\n\n4. DS\n\n5. Back\n\n";
cout << "\nEnter choice: ";
string programChoice;
getline(cin, programChoice);

// Convert to lowercase
for (int i = 0; programChoice[i]; i++) {
if (programChoice[i] >= 'A' && programChoice[i] <= 'Z') {
programChoice[i] += 32;
```




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```
}
}

string selectedProgram;
if (programChoice == "1" || programChoice == "ai") {
    selectedProgram = "AI";
}
else if (programChoice == "2" || programChoice == "cs") {
    selectedProgram = "CS";
}
else if (programChoice == "3" || programChoice == "se") {
    selectedProgram = "SE";
}
else if (programChoice == "4" || programChoice == "ds") {
    selectedProgram = "DS";
}
else if (programChoice == "5" || programChoice == "b" || programChoice == "back") {
    break;
}
else {
    cout << "\n\nInvalid program choice. Press Enter to try again...";
    cin.ignore();
    continue;
}

system("cls");
cout << "\n\n--- Enter Graduate Student Information ---\n\n";

// Name (with validation)
while (true) {
    cout << "\nName: ";
    getline(cin, name[totalStudents]);
    bool valid = true;
    for (int i = 0; name[totalStudents][i]; i++) {
        char c = name[totalStudents][i];
        if (!(c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z') || c == ' ')) {
            valid = false;
            break;
        }
    }
    if (valid && name[totalStudents][0] != '\0') {
        break;
    }
    cout << "\n\nInvalid name. Only alphabets and spaces allowed. Press Enter to try again...";
    cin.ignore();
    system("cls");
    cout << "\n\n--- Enter Graduate Student Information ---\n";
}

// Father's Name (with validation)
while (true) {
    cout << "\nFather's Name: ";
    getline(cin, fatherName[totalStudents]);
    bool valid = true;
```




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```
for (int i = 0; fatherName[totalStudents][i]; i++) {
    char c = fatherName[totalStudents][i];
    if (!((c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z') || c == ' ')) {
        valid = false;
        break;
    }
}
if (valid && fatherName[totalStudents][0] != '\0') {
    break;
}
cout << "\nInvalid father's name. Only alphabets and spaces allowed. Press Enter to try again...";
cin.ignore();
system("cls");
cout << "\n\n--- Enter Graduate Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
}

// CNIC (with validation)
while (true) {
    cout << "\nCNIC (13 digits without dashes): ";
    getline(cin, cnic[totalStudents]);
    bool valid = true;
    int digits = 0;
    for (int i = 0; cnic[totalStudents][i]; i++) {
        if (cnic[totalStudents][i] < '0' || cnic[totalStudents][i] > '9') {
            valid = false;
            break;
        }
        digits++;
    }
    if (valid && digits == 13) {
        break;
    }
    cout << "\nInvalid CNIC. Must be exactly 13 digits. Press Enter to try again...";
    cin.ignore();
    system("cls");
    cout << "\n--- Enter Graduate Student Information ---\n";
    cout << "\nName: " << name[totalStudents] << endl;
    cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
}

// Gender (with validation)
while (true) {
    cout << "\nGender (M/F): ";
    getline(cin, gender[totalStudents]);
    if (gender[totalStudents][0] && !gender[totalStudents][1]) {
        char g = gender[totalStudents][0];
        if (g >= 'A' && g <= 'Z') {
            g += 32;
        }
        if (g == 'm' || g == 'male' || g == 'MALE') {
            gender[totalStudents] = "Male";
            break;
        }
    }
}
```



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```
else if (g == 'f' || g == 'F') {
    gender[totalStudents] = "Female";
    break;
}
}

cout << "\nInvalid gender input. Press Enter to try again...";
cin.ignore();
system("cls");
cout << "\n\n--- Enter Graduate Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
cout << "\nCNIC: " << cnic[totalStudents] << endl;
}

// Address
cout << "\nAddress: ";
getline(cin, address[totalStudents]);

// Age (with validation)
while (true) {
    cout << "\nAge: ";
    string ageInput;
    getline(cin, ageInput);
    bool valid = true;
    for (int i = 0; ageInput[i]; i++) {
        if (ageInput[i] < '0' || ageInput[i] > '9') {
            valid = false;
            break;
        }
    }
    if (valid && ageInput[0] != '\0') {
        age[totalStudents] = 0;
        for (int i = 0; ageInput[i]; i++) {
            age[totalStudents] = age[totalStudents] * 10 + (ageInput[i] - '0');
        }
        if (age[totalStudents] > 0) break;
    }
    cout << "\nInvalid age. Please enter a positive number. Press Enter to continue...";
    cin.ignore();
    system("cls");
    cout << "\n\n--- Enter Graduate Student Information ---\n";
    cout << "\nName: " << name[totalStudents] << endl;
    cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
    cout << "\nCNIC: " << cnic[totalStudents] << endl;
    cout << "\nGender: " << gender[totalStudents] << endl;
    cout << "\nAddress: " << address[totalStudents] << endl;
}

// Contact (with validation)
while (true) {
    cout << "\nContact Number (11 digits): ";
    getline(cin, contact[totalStudents]);
    bool valid = true;
    int digits = 0;
```



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```
for (int i = 0; contact[totalStudents][i]; i++) {
    if (contact[totalStudents][i] < '0' || contact[totalStudents][i] > '9') {
        valid = false;
        break;
    }
    digits++;
}
if (valid && digits == 11) {
    break;
}
cout << "\nInvalid contact number. Must be exactly 11 digits. Press Enter to try again...";
cin.ignore();
system("cls");
cout << "\n\n--- Enter Graduate Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
cout << "\nCNIC: " << cnic[totalStudents] << endl;
cout << "\nGender: " << gender[totalStudents] << endl;
cout << "\nAddress: " << address[totalStudents] << endl;
cout << "\nAge: " << age[totalStudents] << endl;
}

// CGPA validation for graduate
interMarks[totalStudents] = -1; // Not applicable for graduate
while (true) {
    cout << "\nEnter CGPA (0.0-4.0): ";
    string gpaInput;
    getline(cin, gpaInput);
    bool valid = true;
    bool hasDecimal = false;
    for (int i = 0; gpaInput[i]; i++) {
        if ((gpaInput[i] < '0' || gpaInput[i] > '9') && gpaInput[i] != '.') {
            valid = false;
            break;
        }
        if (gpaInput[i] == '.') {
            if (hasDecimal) {
                valid = false;
                break;
            }
            hasDecimal = true;
        }
    }
    if (valid && gpaInput[0] != '\0') {
        gpa[totalStudents] = 0;
        float decimalPlace = 0.1;
        bool afterDecimal = false;
        for (int i = 0; gpaInput[i]; i++) {
            if (gpaInput[i] == '.') {
                afterDecimal = true;
                continue;
            }
            if (!afterDecimal) {
                gpa[totalStudents] = gpa[totalStudents] * 10 + (gpaInput[i] - '0');
```



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```
}
else {
gpa[totalStudents] += (gpaInput[i] - '0') * decimalPlace;
decimalPlace /= 10;
}
}
if (gpa[totalStudents] >= 0.0 && gpa[totalStudents] <= 4.0) break;
}
cout << "\nInvalid CGPA (must be 0.0-4.0). Press Enter to try again...";
cin.ignore();
system("cls");
cout << "\n\n--- Enter Graduate Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
cout << "\nCNIC: " << cnic[totalStudents] << endl;
cout << "\nGender: " << gender[totalStudents] << endl;
cout << "\nAddress: " << address[totalStudents] << endl;
cout << "\nAge: " << age[totalStudents] << endl;
cout << "\nContact: " << contact[totalStudents] << endl;
}

// Hostel (with validation)
while (true) {
cout << "\nRequire Hostel (Y/N/1/0): ";
getline(cin, hostel[totalStudents]);
if (hostel[totalStudents][0] && !hostel[totalStudents][1]) {
char h = hostel[totalStudents][0];
if (h >= 'A' && h <= 'Z') {
h += 32;
}
if (h == 'y' || h == '1' || h == 'yes') {
hostel[totalStudents] = "Yes";
break;
}
else if (h == 'n' || h == '0' || h == 'no') {
hostel[totalStudents] = "No";
break;
}
}
}
cout << "\nInvalid input. Please enter Y/N or 1/0. Press Enter to try again...";
cin.ignore();
system("cls");
cout << "\n\n--- Enter Graduate Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
cout << "\nCNIC: " << cnic[totalStudents] << endl;
cout << "\nGender: " << gender[totalStudents] << endl;
cout << "\nAddress: " << address[totalStudents] << endl;
cout << "\nAge: " << age[totalStudents] << endl;
cout << "\nContact: " << contact[totalStudents] << endl;
cout << "\nCGPA: " << gpa[totalStudents] << endl;
}

// Transport (with validation)
```



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```
while (true) {
    cout << "\nRequire Transport (Y/N/1/0): ";
    getline(cin, transport[totalStudents]);
    if (transport[totalStudents][0] && !transport[totalStudents][1]) {
        char t = transport[totalStudents][0];
        if (t >= 'A' && t <= 'Z') {
            t += 32;
        }
        if (t == 'y' || t == '1' || t == 'yes' || t == 'YES') {
            transport[totalStudents] = "Yes";
            break;
        }
        else if (t == 'n' || t == '0' || t == 'no' || t == 'NO') {
            transport[totalStudents] = "No";
            break;
        }
        cout << "\nInvalid input. Please enter Y/N or 1/0. Press Enter to try again...";
        cin.ignore();
        system("cls");
        cout << "\n--- Enter Graduate Student Information ---\n";
        cout << "\nName: " << name[totalStudents] << endl;
        cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
        cout << "\nCNIC: " << cnic[totalStudents] << endl;
        cout << "\nGender: " << gender[totalStudents] << endl;
        cout << "\nAddress: " << address[totalStudents] << endl;
        cout << "\nAge: " << age[totalStudents] << endl;
        cout << "\nContact: " << contact[totalStudents] << endl;
        cout << "\nCGPA: " << gpa[totalStudents] << endl;
        cout << "\nHostel: " << hostel[totalStudents] << endl;
    }

    // Assign remaining fields
    studentID[totalStudents] = idCounter++;
    program[totalStudents] = selectedProgram;
    level[totalStudents] = selectedLevel;
    semester[totalStudents] = "1st";
    department[totalStudents] = "FOIT";
    field[totalStudents] = selectedProgram;

    totalStudents++;

    cout << "\n\nGraduate student added successfully!\n";
    cout << "\nPress Enter to return to Main Menu...";
    cin.ignore();
    break;
}
break;
}
else if (levelChoice == "3" || levelChoice == "p" || levelChoice == "postgraduate") {
    string selectedLevel = "Postgraduate";
    while (true) {
        system("cls");
        cout << "\n\n-- Postgraduate Program Selection --\n";
```



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```
cout << "1. AI\n\n2. CS\n\n3. SE\n\n4. DS\n\n5. Back\n\n";
cout << "\nEnter choice: ";
string programChoice;
getline(cin, programChoice);

// Convert to lowercase
for (int i = 0; programChoice[i]; i++) {
    if (programChoice[i] >= 'A' && programChoice[i] <= 'Z') {
        programChoice[i] += 32;
    }
}

string selectedProgram;
if (programChoice == "1" || programChoice == "ai") {
    selectedProgram = "AI";
}
else if (programChoice == "2" || programChoice == "cs") {
    selectedProgram = "CS";
}
else if (programChoice == "3" || programChoice == "se") {
    selectedProgram = "SE";
}
else if (programChoice == "4" || programChoice == "ds") {
    selectedProgram = "DS";
}
else if (programChoice == "5" || programChoice == "b" || programChoice == "back") {
    break;
}
else {
    cout << "\nInvalid program choice. Press Enter to try again...";
    cin.ignore();
    continue;
}

system("cls");
cout << "\n\n--- Enter Postgraduate Student Information ---\n";

// Name (with validation)
while (true) {
    cout << "\nName: ";
    getline(cin, name[totalStudents]);
    bool valid = true;
    for (int i = 0; name[totalStudents][i]; i++) {
        char c = name[totalStudents][i];
        if (!(c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z') || c == ' ')) {
            valid = false;
            break;
        }
    }
    if (valid && name[totalStudents][0] != '\0') {
        break;
    }
    cout << "\nInvalid name. Only alphabets and spaces allowed. Press Enter to try again...";
    cin.ignore();
}
```



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```
system("cls");
cout << "\n\n--- Enter Postgraduate Student Information ---\n";
}

// Father's Name (with validation)
while (true) {
    cout << "\nFather's Name: ";
    getline(cin, fatherName[totalStudents]);
    bool valid = true;
    for (int i = 0; fatherName[totalStudents][i]; i++) {
        char c = fatherName[totalStudents][i];
        if (!((c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z') || c == ' ')) {
            valid = false;
            break;
        }
    }
    if (valid && fatherName[totalStudents][0] != '\0') {
        break;
    }
    cout << "\nInvalid father's name. Only alphabets and spaces allowed. Press Enter to try again...";
    cin.ignore();
    system("cls");
    cout << "\n\n--- Enter Postgraduate Student Information ---\n";
    cout << "\nName: " << name[totalStudents] << endl;
}

// CNIC (with validation)
while (true) {
    cout << "\nCNIC (13 digits without dashes): ";
    getline(cin, cnic[totalStudents]);
    bool valid = true;
    int digits = 0;
    for (int i = 0; cnic[totalStudents][i]; i++) {
        if (cnic[totalStudents][i] < '0' || cnic[totalStudents][i] > '9') {
            valid = false;
            break;
        }
    }
    digits++;
    if (valid && digits == 13) {
        break;
    }
    cout << "\nInvalid CNIC. Must be exactly 13 digits. Press Enter to try again...";
    cin.ignore();
    system("cls");
    cout << "\n\n--- Enter Postgraduate Student Information ---\n";
    cout << "\nName: " << name[totalStudents] << endl;
    cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
}

// Gender (with validation)
while (true) {
    cout << "\nGender (M/F): ";
    getline(cin, gender[totalStudents]);
}
```




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```
if (gender[totalStudents][0] && !gender[totalStudents][1]) {
char g = gender[totalStudents][0];
if (g >= 'A' && g <= 'Z') {
g += 32;
}
if (g == 'm' || g == 'male' || g == 'MALE') {
gender[totalStudents] = "Male";
break;
}
else if (g == 'f') {
gender[totalStudents] = "Female";
break;
}
}
cout << "\nInvalid gender input. Press Enter to try again...";
cin.ignore();
system("cls");
cout << "\n\n--- Enter Postgraduate Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
cout << "\nCNIC: " << cnic[totalStudents] << endl;
}

// Address
cout << "\nAddress: ";
getline(cin, address[totalStudents]);

// Age (with validation)
while (true) {
cout << "\nAge: ";
string ageInput;
getline(cin, ageInput);
bool valid = true;
for (int i = 0; ageInput[i]; i++) {
if (ageInput[i] < '0' || ageInput[i] > '9') {
valid = false;
break;
}
}
if (valid && ageInput[0] != '\0') {
age[totalStudents] = 0;
for (int i = 0; ageInput[i]; i++) {
age[totalStudents] = age[totalStudents] * 10 + (ageInput[i] - '0');
}
if (age[totalStudents] > 0) break;
}
cout << "\nInvalid age. Please enter a positive number. Press Enter to continue...";
cin.ignore();
system("cls");
cout << "\n\n--- Enter Postgraduate Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
cout << "\nCNIC: " << cnic[totalStudents] << endl;
cout << "\nGender: " << gender[totalStudents] << endl;
```




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```
cout << "\nAddress: " << address[totalStudents] << endl;
}

// Contact (with validation)
while (true) {
    cout << "\nContact Number (11 digits): ";
    getline(cin, contact[totalStudents]);
    bool valid = true;
    int digits = 0;
    for (int i = 0; contact[totalStudents][i]; i++) {
        if (contact[totalStudents][i] < '0' || contact[totalStudents][i] > '9') {
            valid = false;
            break;
        }
        digits++;
    }
    if (valid && digits == 11) {
        break;
    }
    cout << "\nInvalid contact number. Must be exactly 11 digits. Press Enter to try again...";
    cin.ignore();
    system("cls");
    cout << "\n\n--- Enter Postgraduate Student Information ---\n";
    cout << "\nName: " << name[totalStudents] << endl;
    cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
    cout << "\nCNIC: " << cnic[totalStudents] << endl;
    cout << "\nGender: " << gender[totalStudents] << endl;
    cout << "\nAddress: " << address[totalStudents] << endl;
    cout << "\nAge: " << age[totalStudents] << endl;
}

// CGPA validation for postgraduate
interMarks[totalStudents] = -1; // Not applicable for postgraduate
while (true) {
    cout << "\nEnter CGPA (0.0-4.0): ";
    string gpaInput;
    getline(cin, gpaInput);
    bool valid = true;
    bool hasDecimal = false;
    for (int i = 0; gpaInput[i]; i++) {
        if ((gpaInput[i] < '0' || gpaInput[i] > '9') && gpaInput[i] != '.') {
            valid = false;
            break;
        }
        if (gpaInput[i] == '.') {
            if (hasDecimal) {
                valid = false;
                break;
            }
            hasDecimal = true;
        }
    }
    if (valid && gpaInput[0] != '\0') {
        gpa[totalStudents] = 0;
    }
}
```



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```
float decimalPlace = 0.1;
bool afterDecimal = false;
for (int i = 0; gpaInput[i]; i++) {
    if (gpaInput[i] == '.') {
        afterDecimal = true;
        continue;
    }
    if (!afterDecimal) {
        gpa[totalStudents] = gpa[totalStudents] * 10 + (gpaInput[i] - '0');
    }
    else {
        gpa[totalStudents] += (gpaInput[i] - '0') * decimalPlace;
        decimalPlace /= 10;
    }
}
if (gpa[totalStudents] >= 0.0 && gpa[totalStudents] <= 4.0) break;
}
cout << "\nInvalid CGPA (must be 0.0-4.0). Press Enter to try again...";
cin.ignore();
system("cls");
cout << "\n\n--- Enter Postgraduate Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
cout << "\nCNIC: " << cnic[totalStudents] << endl;
cout << "\nGender: " << gender[totalStudents] << endl;
cout << "\nAddress: " << address[totalStudents] << endl;
cout << "\nAge: " << age[totalStudents] << endl;
cout << "\nContact: " << contact[totalStudents] << endl;
}

// Hostel (with validation)
while (true) {
    cout << "\nRequire Hostel (Y/N/1/0): ";
    getline(cin, hostel[totalStudents]);
    if (hostel[totalStudents][0] && !hostel[totalStudents][1]) {
        char h = hostel[totalStudents][0];
        if (h >= 'A' && h <= 'Z') {
            h += 32;
        }
        if (h == 'y' || h == '1') {
            hostel[totalStudents] = "Yes";
            break;
        }
        else if (h == 'n' || h == '0') {
            hostel[totalStudents] = "No";
            break;
        }
    }
}
cout << "\nInvalid input. Please enter Y/N or 1/0. Press Enter to try again...";
cin.ignore();
system("cls");
cout << "\n\n--- Enter Postgraduate Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
```



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```
cout << "\nCNIC: " << cnic[totalStudents] << endl;
cout << "\nGender: " << gender[totalStudents] << endl;
cout << "\nAddress: " << address[totalStudents] << endl;
cout << "\nAge: " << age[totalStudents] << endl;
cout << "\nContact: " << contact[totalStudents] << endl;
cout << "\nCGPA: " << gpa[totalStudents] << endl;
}

// Transport (with validation)
while (true) {
cout << "\nRequire Transport (Y/N/1/0): ";
getline(cin, transport[totalStudents]);
if (transport[totalStudents][0] && !transport[totalStudents][1]) {
char t = transport[totalStudents][0];
if (t >= 'A' && t <= 'Z') {
t += 32;
}
if (t == 'y' || t == '1' || t == 'yes') {
transport[totalStudents] = "Yes";
break;
}
else if (t == 'n' || t == '0' || t == 'no') {
transport[totalStudents] = "No";
break;
}
}
cout << "\nInvalid input. Please enter Y/N or 1/0. Press Enter to try again...";
cin.ignore();
system("cls");
cout << "\n--- Enter Postgraduate Student Information ---\n";
cout << "\nName: " << name[totalStudents] << endl;
cout << "\nFather's Name: " << fatherName[totalStudents] << endl;
cout << "\nCNIC: " << cnic[totalStudents] << endl;
cout << "\nGender: " << gender[totalStudents] << endl;
cout << "\nAddress: " << address[totalStudents] << endl;
cout << "\nAge: " << age[totalStudents] << endl;
cout << "\nContact: " << contact[totalStudents] << endl;
cout << "\nCGPA: " << gpa[totalStudents] << endl;
cout << "\nHostel: " << hostel[totalStudents] << endl;
}

// Assign remaining fields
studentID[totalStudents] = idCounter++;
program[totalStudents] = selectedProgram;
level[totalStudents] = selectedLevel;
semester[totalStudents] = "1st";
department[totalStudents] = "FOIT";
field[totalStudents] = selectedProgram;

totalStudents++;

cout << "\nPostgraduate student added successfully!\n";
cout << "\nPress Enter to return to Main Menu...";
cin.ignore();
```



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```
break;
}
break;
}
else if (levelChoice == "4" || levelChoice == "b" || levelChoice == "back") {
break;
}
else {
cout << "\nInvalid choice. Press Enter to try again...";
cin.ignore();
}
}
}

else if (input == "2" || input == "a" || input == "admin") {
system("cls");
string adminID, adminPass;
cout << "Enter Admin ID: ";
getline(cin, adminID);
cout << "Enter Password: ";
getline(cin, adminPass);

if (adminID == "FAB4" && adminPass == "1234") {
while (true) {
system("cls");
cout << "\n-- Admin Panel --\n";
cout << "\n1. Display All Students\n";
cout << "\n2. Search by Name\n";
cout << "\n3. Update Student\n";
cout << "\n4. Remove Student\n";
cout << "\n5. Back to Main Menu\n";
cout << "\n\nEnter choice: ";
string adminChoice;
getline(cin, adminChoice);

if (adminChoice == "1") {
system("cls");
if (totalStudents == 0) {
cout << "\nNo student records available.\n";
}
else {
cout << "\n" << string(100, '-') << "\n";
cout << left << setw(10) << "ID" << setw(20) << "Name" << setw(10) << "Semester"
<< setw(10) << "Dept" << setw(10) << "Field" << setw(15) << "Level"
<< setw(15) << "Scholarship" << "\n";
cout << string(100, '-') << "\n";

for (int i = 0; i < totalStudents; i++) {
cout << left << setw(10) << studentID[i]
<< setw(20) << name[i].substr(0, 20)
<< setw(10) << semester[i]
<< setw(10) << department[i]
<< setw(10) << field[i]
<< setw(15) << level[i];
```



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```
string scholarship;
if (level[i] == "Undergraduate") {
float perc = (interMarks[i] / 1200.0f) * 100;
if (perc >= 85) scholarship = "100%";
else if (perc >= 80) scholarship = "75%";
else if (perc >= 75) scholarship = "50%";
else if (perc >= 70) scholarship = "25%";
else scholarship = "None";
}
else {
if (gpa[i] >= 4.0) scholarship = "100%";
else if (gpa[i] >= 3.75) scholarship = "75%";
else if (gpa[i] >= 3.5) scholarship = "50%";
else if (gpa[i] >= 3.2) scholarship = "25%";
else scholarship = "None";
}

cout << setw(15) << scholarship << endl;
}
cout << string(100, '-') << "\n";
}
cout << "\n\nPress Enter to return to Admin Panel...";
cin.ignore();
}
else if (adminChoice == "2") {
cout << "\n\nEnter student name to search: ";
string searchName;
getline(cin, searchName);
bool found = false;
for (int i = 0; i < totalStudents; i++) {
// Convert both to lowercase for case-insensitive search
bool match = true;
for (int j = 0; searchName[j] && name[i][j]; j++) {
char c1 = searchName[j];
char c2 = name[i][j];
if (c1 >= 'A' && c1 <= 'Z') c1 += 32;
if (c2 >= 'A' && c2 <= 'Z') c2 += 32;
if (c1 != c2) {
match = false;
break;
}
}
if (match) {
found = true;
cout << "\nStudent Found:\n";
cout << "\nID: " << studentID[i] << "\n";
cout << "\nName: " << name[i] << "\n";
cout << "\nFather's Name: " << fatherName[i] << "\n";
cout << "\nCNIC: " << cnic[i] << "\n";
cout << "\nGender: " << gender[i] << "\n";
cout << "\nAddress: " << address[i] << "\n";
cout << "\nAge: " << age[i] << "\n";
cout << "\nContact: " << contact[i] << "\n";
cout << "\nLevel: " << level[i] << "\n";
```



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```
cout << "\nProgram: " << program[i] << "\n";
cout << "\nField: " << field[i] << "\n";
cout << "\nDepartment: " << department[i] << "\n";
cout << "\nSemester: " << semester[i] << "\n";
if (level[i] == "Undergraduate") {
cout << "Intermediate Marks: " << interMarks[i] << "/1200\n";
}
else {
cout << "\nCGPA: " << gpa[i] << "/4.0\n";
}
cout << "\nHostel: " << hostel[i] << "\n";
cout << "\nTransport: " << transport[i] << "\n";
break;
}
}
if (!found) cout << "\n\nStudent not found.\n";
cout << "\n\nPress Enter to return to Admin Panel...";
cin.ignore();
}
else if (adminChoice == "3") {
// Update student
cout << "\n\nEnter student ID to update: ";
string idInput;
getline(cin, idInput);
bool validID = true;
for (int i = 0; idInput[i]; i++) {
if (idInput[i] < '0' || idInput[i] > '9') {
validID = false;
break;
}
}
if (validID && idInput[0] != '\0') {
int id = 0;
for (int i = 0; idInput[i]; i++) {
id = id * 10 + (idInput[i] - '0');
}
int index = -1;
for (int i = 0; i < totalStudents; i++) {
if (studentID[i] == id) {
index = i;
break;
}
}
if (index == -1) {
cout << "\nStudent not found with ID: " << id << endl;
}
else {
cout << "\n\nCurrent student information:\n";
cout << "\n1. Name: " << name[index] << endl;
cout << "\n2. Father's Name: " << fatherName[index] << endl;
cout << "\n3. CNIC: " << cnic[index] << endl;
cout << "\n4. Gender: " << gender[index] << endl;
cout << "\n5. Address: " << address[index] << endl;
cout << "\n6. Age: " << age[index] << endl;
```



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```
cout << "\n7. Contact: " << contact[index] << endl;
if (level[index] == "Undergraduate") {
cout << "\n8. Intermediate Marks: " << interMarks[index] << endl;
}
else {
cout << "\n8. CGPA: " << gpa[index] << endl;
}
cout << "\n9. Hostel: " << hostel[index] << endl;
cout << "\n10. Transport: " << transport[index] << endl;

cout << "\n\nEnter field number to update (1-10) or 0 to cancel: ";
string fieldChoice;
getline(cin, fieldChoice);
bool validChoice = true;
for (int i = 0; fieldChoice[i]; i++) {
if (fieldChoice[i] < '0' || fieldChoice[i] > '9') {
validChoice = false;
break;
}
}
if (validChoice && fieldChoice[0] != '\0') {
int choice = 0;
for (int i = 0; fieldChoice[i]; i++) {
choice = choice * 10 + (fieldChoice[i] - '0');
}
if (choice == 0) {
cout << "\nUpdate cancelled.\n";
}
else if (choice >= 1 && choice <= 10) {
cout << "\nEnter new value: ";
string newValue;
getline(cin, newValue);

switch (choice) {
case 1:
{
bool valid = true;
for (int i = 0; newValue[i]; i++) {
char c = newValue[i];
if (!(c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z') || c == ' ') {
valid = false;
break;
}
}
if (valid && newValue[0] != '\0') {
name[index] = newValue;
cout << "\nName updated successfully.\n";
}
else {
cout << "\nInvalid name. Only alphabets and spaces allowed. Update failed.\n";
}
}
break;
case 2:
```




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```
{
bool valid = true;
for (int i = 0; newValue[i]; i++) {
char c = newValue[i];
if (!(c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z') || c == ' ')) {
valid = false;
break;
}
}
if (valid && newValue[0] != '\0') {
fatherName[index] = newValue;
cout << "\nFather's name updated successfully.\n";
}
else {
cout << "\nInvalid father's name. Only alphabets and spaces allowed. Update failed.\n";
}
}
break;
case 3:
{
bool valid = true;
int digits = 0;
for (int i = 0; newValue[i]; i++) {
if (newValue[i] < '0' || newValue[i] > '9') {
valid = false;
break;
}
}
digits++;
}
if (valid && digits == 13) {
cnic[index] = newValue;
cout << "\nCNIC updated successfully.\n";
}
else {
cout << "\nInvalid CNIC. Must be exactly 13 digits. Update failed.\n";
}
}
break;
case 4:
if (newValue[0] && !newValue[1]) {
char g = newValue[0];
if (g >= 'A' && g <= 'Z') {
g += 32;
}
if (g == 'm') {
gender[index] = "Male";
}
else if (g == 'f') {
gender[index] = "Female";
}
else {
cout << "\nInvalid gender. Update failed.\n";
break;
}
}
```




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```
cout << "\nGender updated successfully.\n";
}
else {
cout << "\nInvalid gender. Update failed.\n";
}
break;
case 5: address[index] = newValue; break;
case 6: {
bool valid = true;
for (int i = 0; newValue[i]; i++) {
if (newValue[i] < '0' || newValue[i] > '9') {
valid = false;
break;
}
}
if (valid && newValue[0] != '\0') {
age[index] = 0;
for (int i = 0; newValue[i]; i++) {
age[index] = age[index] * 10 + (newValue[i] - '0');
}
cout << "\nAge updated successfully.\n";
}
else {
cout << "\nInvalid age. Update failed.\n";
}
break;
}
case 7:
{
bool valid = true;
int digits = 0;
for (int i = 0; newValue[i]; i++) {
if (newValue[i] < '0' || newValue[i] > '9') {
valid = false;
break;
}
digits++;
}
if (valid && digits == 11) {
contact[index] = newValue;
cout << "\nContact updated successfully.\n";
}
else {
cout << "\nInvalid contact. Must be exactly 11 digits. Update failed.\n";
}
}
break;
case 8:
if (level[index] == "Undergraduate") {
bool valid = true;
for (int i = 0; newValue[i]; i++) {
if (newValue[i] < '0' || newValue[i] > '9') {
valid = false;
break;
}
```



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```
}
}
if (valid && newValue[0] != '\0') {
    interMarks[index] = 0;
    for (int i = 0; newValue[i]; i++) {
        interMarks[index] = interMarks[index] * 10 + (newValue[i] - '0');
    }
    if (interMarks[index] >= 0 && interMarks[index] <= 1200) {
        cout << "\nMarks updated successfully.\n";
    }
    else {
        cout << "\nMarks must be between 0-1200. Update failed.\n";
    }
}
else {
    cout << "\nInvalid marks. Update failed.\n";
}
}
else {
    bool valid = true;
    bool hasDecimal = false;
    for (int i = 0; newValue[i]; i++) {
        if ((newValue[i] < '0' || newValue[i] > '9') && newValue[i] != '.') {
            valid = false;
            break;
        }
        if (newValue[i] == '.') {
            if (hasDecimal) {
                valid = false;
                break;
            }
            hasDecimal = true;
        }
    }
    if (valid && newValue[0] != '\0') {
        gpa[index] = 0;
        float decimalPlace = 0.1;
        bool afterDecimal = false;
        for (int i = 0; newValue[i]; i++) {
            if (newValue[i] == '.') {
                afterDecimal = true;
                continue;
            }
            if (!afterDecimal) {
                gpa[index] = gpa[index] * 10 + (newValue[i] - '0');
            }
            else {
                gpa[index] += (newValue[i] - '0') * decimalPlace;
                decimalPlace /= 10;
            }
        }
        if (gpa[index] >= 0.0 && gpa[index] <= 4.0) {
            cout << "\nCGPA updated successfully.\n";
        }
    }
}
```



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```
else {
cout << "\nCGPA must be between 0.0-4.0. Update failed.\n";
}
}
else {
cout << "\nInvalid CGPA. Update failed.\n";
}
}
break;
case 9:
if (newValue[0] && !newValue[1]) {
char h = newValue[0];
if (h >= 'A' && h <= 'Z') {
h += 32;
}
if (h == 'y' || h == '1') {
hostel[index] = "Yes";
cout << "\nHostel requirement updated successfully.\n";
}
else if (h == 'n' || h == '0') {
hostel[index] = "No";
cout << "\nHostel requirement updated successfully.\n";
}
else {
cout << "\nInvalid input. Update failed.\n";
}
}
else {
cout << "\nInvalid input. Update failed.\n";
}
}
break;
case 10:
if (newValue[0] && !newValue[1]) {
char t = newValue[0];
if (t >= 'A' && t <= 'Z') {
t += 32;
}
if (t == 'y' || t == '1') {
transport[index] = "Yes";
cout << "\nTransport requirement updated successfully.\n";
}
else if (t == 'n' || t == '0') {
transport[index] = "No";
cout << "\nTransport requirement updated successfully.\n";
}
else {
cout << "\nInvalid input. Update failed.\n";
}
}
else {
cout << "\nInvalid input. Update failed.\n";
}
}
break;
}
```



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```
}
else {
cout << "\nInvalid choice. Please enter 0-10.\n";
}
}
else {
cout << "\nInvalid choice. Please enter a number.\n";
}
}
}
else {
cout << "\nInvalid ID. Please enter a number.\n";
}
cout << "\nPress Enter to return to Admin Panel...";
cin.ignore();
}
else if (adminChoice == "4") {
// Remove student
cout << "\n\nEnter student ID to remove: ";
string idInput;
getline(cin, idInput);
bool validID = true;
for (int i = 0; idInput[i]; i++) {
if (idInput[i] < '0' || idInput[i] > '9') {
validID = false;
break;
}
}
if (validID && idInput[0] != '\0') {
int id = 0;
for (int i = 0; idInput[i]; i++) {
id = id * 10 + (idInput[i] - '0');
}
int index = -1;
for (int i = 0; i < totalStudents; i++) {
if (studentID[i] == id) {
index = i;
break;
}
}
if (index == -1) {
cout << "\nStudent not found with ID: " << id << endl;
}
else {
cout << "\nAre you sure you want to remove " << name[index] << " (ID: " << id << ")? (Y/N): ";
string confirm;
getline(cin, confirm);
if (confirm[0] && !confirm[1]) {
char c = confirm[0];
if (c >= 'A' && c <= 'Z') {
c += 32;
}
if (c == 'y') {
// Shift all elements after index to the left
```



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```
for (int i = index; i < totalStudents - 1; i++) {
name[i] = name[i + 1];
fatherName[i] = fatherName[i + 1];
cnic[i] = cnic[i + 1];
gender[i] = gender[i + 1];
address[i] = address[i + 1];
contact[i] = contact[i + 1];
age[i] = age[i + 1];
interMarks[i] = interMarks[i + 1];
gpa[i] = gpa[i + 1];
hostel[i] = hostel[i + 1];
transport[i] = transport[i + 1];
studentID[i] = studentID[i + 1];
program[i] = program[i + 1];
level[i] = level[i + 1];
semester[i] = semester[i + 1];
department[i] = department[i + 1];
field[i] = field[i + 1];
}
totalStudents--;
cout << "\nStudent removed successfully.\n";
}
else {
cout << "\nRemoval cancelled.\n";
}
}
else {
cout << "\nInvalid input. Removal cancelled.\n";
}
}
}
else {
cout << "\nInvalid ID. Please enter a number.\n";
}
cout << "\n\nPress Enter to return to Admin Panel...";
cin.ignore();
}
else if (adminChoice == "5") {
break;
}
else {
cout << "\nInvalid admin panel choice. Press Enter to try again...";
cin.ignore();
}
}
}
else {
cout << "\nIncorrect admin input. Press Enter to return to Main Menu...";
cin.ignore();
cin.get();
}
}
else if (input == "3" || input == "e" || input == "exit") {
cout << "\nExiting program...\n";
```



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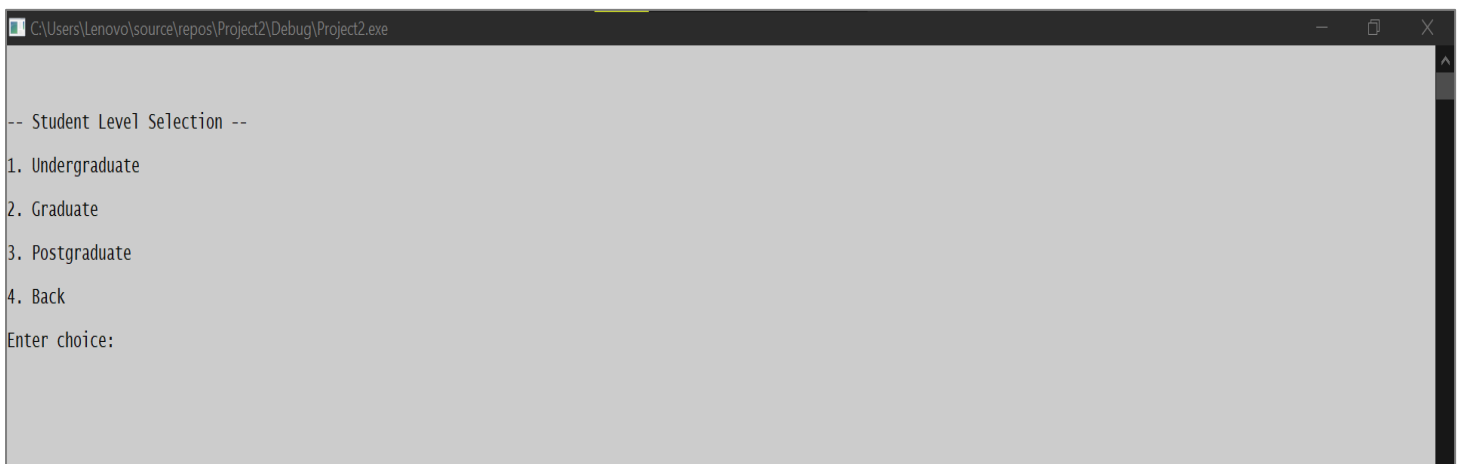
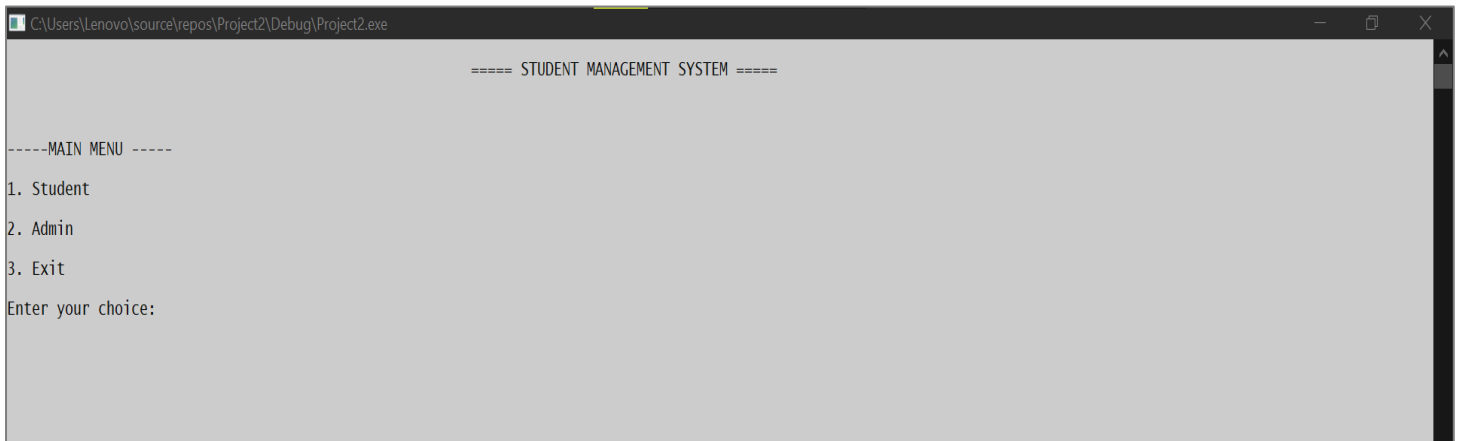
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```
break;
}
else {
cout << "\nInvalid main menu choice. Press Enter to try again...";
cin.ignore();
cin.get();
}
}

return 0;
}
```

Program Menu :





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```
C:\Users\Lenovo\source\repos\Project2\Debug\Project2.exe

-- Program Selection --
1. AI
2. CS
3. SE
4. DS
5. Back
Enter choice:
```

```
C:\Users\Lenovo\source\repos\Project2\Debug\Project2.exe

-- Admin Panel --
1. Display All Students
2. Search by Name
3. Update Student
4. Remove Student
5. Back to Main Menu
Enter choice:
```



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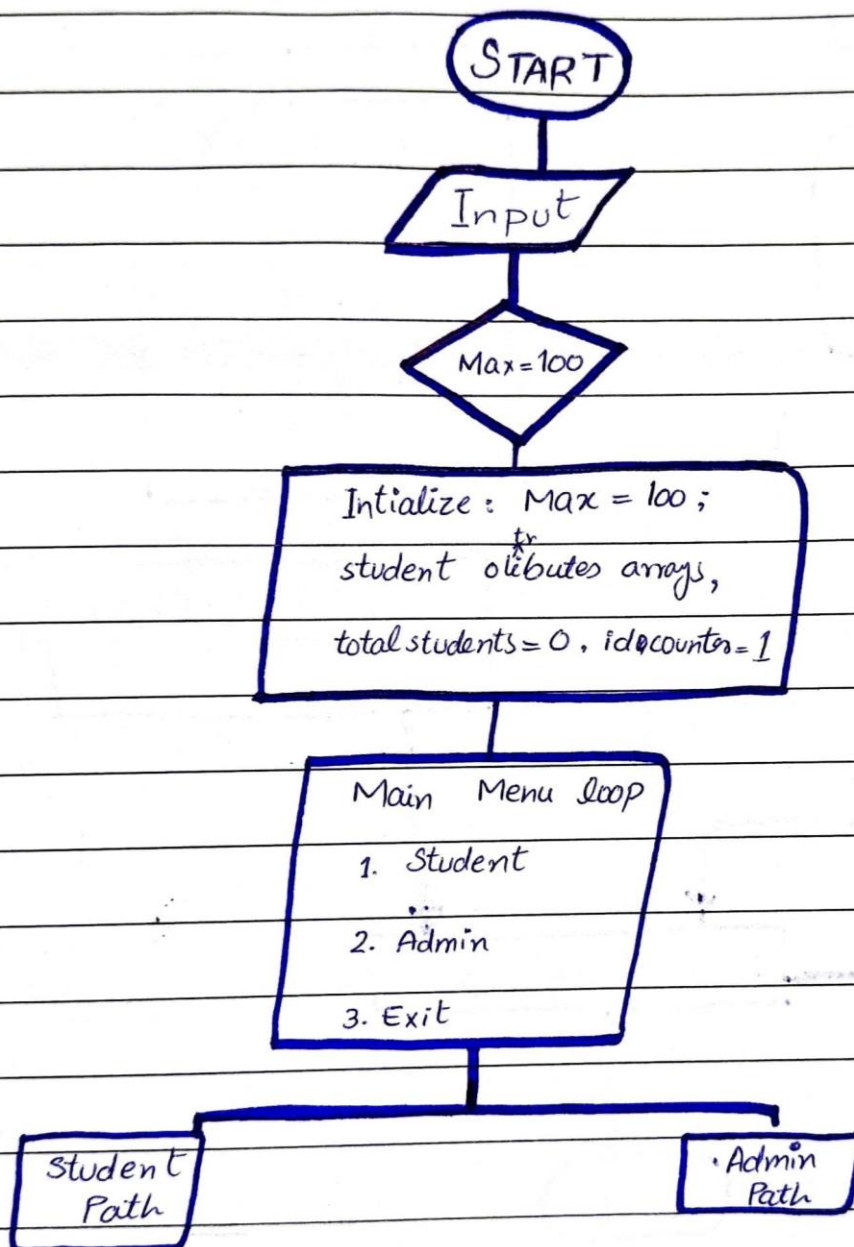
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FlowChart :

INTRODUCTION TO COMPUTING

PROJECT PRESENTATION "STUDENT MANAGEMENT SYSTEM"

FLOW CHART





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